

The empirical recurrence for OEIS A316957, recovered from its tail

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Abstract

OEIS A316957 records “Empirical recurrence of order 58” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 116$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 60$. Its last coefficient is $c_{58} = 92362 \neq 0$, so no further tail satisfies anything shorter; any order-58 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316957 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 1, 2, 3, 4, 5 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 512, 10243, 194917, 3796326, 73529753, 1426910760, 27687032205, ...

The entry states:

Empirical recurrence of order 58 (see link above)

As of the “Last modified” line on the live entry (Jul 17 2018 12:31:28, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 116$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 116$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 58: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 58.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 232$ exact terms from $n = 2$ on, it returns order 58 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{58} c_i a(n-i),$$

c_1	17	c_{16}	−4610331114	c_{31}	−11656418203	c_{46}	−1134250437
c_2	152	c_{17}	475946154	c_{32}	−38646545782	c_{47}	−1348621460
c_3	−1429	c_{18}	12158033499	c_{33}	−33948503528	c_{48}	−239193249
c_4	−13839	c_{19}	19287540204	c_{34}	5813644954	c_{49}	−168378061
c_5	13430	c_{20}	9876680253	c_{35}	53816750868	c_{50}	−155449167
c_6	419345	c_{21}	−14608895298	c_{36}	52118391024	c_{51}	18931395
c_7	1088790	c_{22}	−27456439236	c_{37}	28758208155	c_{52}	−4329821
c_8	−3502012	c_{23}	−12156271057	c_{38}	8396731723	c_{53}	−7269510
c_9	−21614698	c_{24}	17509929604	c_{39}	−10961891312	c_{54}	11650851
c_{10}	−21651758	c_{25}	33236469768	c_{40}	−8937515257	c_{55}	−3110720
c_{11}	100277434	c_{26}	20782619354	c_{41}	−3690704486	c_{56}	−3672163
c_{12}	369595997	c_{27}	−15640649980	c_{42}	2006971929	c_{57}	1040062
c_{13}	376634353	c_{28}	−48971445125	c_{43}	2979351756	c_{58}	92362
c_{14}	−876847580	c_{29}	−51537062563	c_{44}	−986613464		
c_{15}	−3414178312	c_{30}	−21391882928	c_{45}	−637544783		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 60$, and it is the recurrence OEIS A316957 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{58} - c_1 x^{57} - \dots - c_{58}$. Its constant term is $-c_{58} = -92362$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 58 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 58, so $\deg q = 58$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 58, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 58, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 92362.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-58 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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